

NY-lite Calibration Weekly Update

Public NYISO targets, ACOPF results, and current feasibility blockers

NPCC / NY-lite Calibration Workstream

July 1, 2026

Executive Summary

- Built the public NYISO calibration scaffold using P-58C zonal loads and P-32 interface flows.
- Preserved the original NPCC New York load scale: **10,902.22 MW**.
- Generated six scaled public scenarios, 66 zonal load target rows, and 30 interface target rows.
- Ran IPOPT ACOPF for original topology and Gilboa-Leeds topology across all six scenarios.
- Current blocker: **all 12 public-load ACOPF runs returned `opf_success = 0`.**

Main interpretation

The public NYISO load shape is a large spatial perturbation from the original NPCC case, especially because NPCC has zero explicit original load in HUD VL and NYC proxy zones.

Completed This Week

- Confirmed S0/original NPCC AC power flow converges but embedded dispatch is not active-power feasible.
- Built S1 from ACOPF solution instead of embedded S0 dispatch.
- Verified S1 original and S1 plus Gilboa-Leeds with IPOPT on the original S1 load point.
- Added public NYISO scenario ingestion and target generation scripts.
- Added IPOPT public-scenario runner with per-interface residual reporting.
- Added Gilboa-Leeds one-dimensional X grid-search driver and smoke-tested it.

Calibration Workflow Implemented

Public data

P-58C zonal load
P-32 interface flow

Scale

Preserve NPCC NY total:
10,902.22 MW

Solve

Run IPOPT ACOPF:
original and GL topology

Fit

Grid search over
 X_{GL}

Public data → scaled targets → NY-lite ACOPF → interface residuals

- Load targets are written as MW targets only, avoiding mixed MW/share inputs.
- P-32 interface targets are used for reporting/objective terms, not as branch ratings.
- The first calibration variable is the Gilboa-Leeds reactance, X_{GL} .

Public Scenario Targets

Scenario	NYISO total MW	NPCC NY MW	Scale γ
Summer peak	30,644.99	10,902.22	0.35576
Winter peak	23,521.35	10,902.22	0.46350
Shoulder light load	11,060.56	10,902.22	0.98568
High NYC plus LI	30,414.51	10,902.22	0.35845
High Total East flow	28,333.18	10,902.22	0.38479
Low Total East flow	23,862.11	10,902.22	0.45688

Scaling rule

$$P_z^{target}(t) = P_{NY}^{NPCC} \frac{L_z^{NYISO}(t)}{\sum_{k=A}^K L_k^{NYISO}(t)} = \gamma(t) L_z^{NYISO}(t)$$

Load Distribution Mismatch

Zone	Name	NPCC share	NYISO avg share	Delta pp
A	WEST	24.04%	9.15%	+14.89
C	CENTRL	16.40%	8.64%	+7.76
D	NORTH	8.05%	2.42%	+5.63
G	HUD VL	0.00%	6.73%	-6.73
I	DUNWOD	21.21%	4.36%	+16.85
J	N.Y.C.	0.00%	34.57%	-34.57
K	LONGIL	7.42%	15.12%	-7.70

Original NPCC

A-F upstate share: **70.29%**
G-K downstate share: **29.71%**

Public NYISO average

A-F upstate share: **37.20%**
G-K downstate share: **62.80%**

Important Mapping Clarification

- HUD VL and NYC proxy buses do exist in the reduced NPCC mapping.
- They have **zero original active load** in the base NPCC case.
- CE UG is a bus, not a zone:
 - Bus 78: CE UG
 - Mapped zone: I / DUNWOD
 - Original load: 2,311.84 MW

Bus	Name	Mapped zone	Base Pd MW
39, 73, 75, 76	Leeds / Pleasant Valley / Ramapo	G / HUD VL	0.00
79, 81, 82	RAV A-3 / Goethals / AK-3	J / N.Y.C.	0.00
78	CE UG	I / DUNWOD	2,311.84

Public ACOPF Results

- Ran six public scenarios for two topology variants:
 - S1 original topology
 - S1 plus Gilboa-Leeds
- Solver/settings:
 - `opf.ac.solver = IPOPT`
 - `opf.flow_lim = S`
 - `opf.violation = 1e-6`
 - `opf.use_vg = 0`
 - `opf.ignore_angle_lim = 0`
- Result: 0 / 12 public ACOPF runs certified feasible.
- The Gilboa-Leeds grid-smoke driver ran, but every trial had `success_count = 0`.

Failure Evidence

- Representative public-load diagnostic:
 - Base constraints: failed.
 - Branch ratings removed only: still failed.
 - Branch ratings plus voltage bounds relaxed: solved, but voltages reached unrealistic levels up to 1.5 pu.
- This points to a combined feasibility issue, with voltage/reactive feasibility as the leading suspect.
- Branch 190 and several ratings remain tight, but branch limits alone do not explain the failure.

Current assessment

The public load-shape shift appears too severe for the current reduced topology, reactive support model, and voltage constraints.

Largest Interface Residuals

Scenario	Variant	Interface	Target MW	Lite MW	Residual MW
Low Total East	GL	Total East	-325.57	2,493.54	2,819.10
Low Total East	Original	Total East	-325.57	2,290.27	2,615.83
Low Total East	GL	Central East	-48.92	2,169.24	2,218.16
Low Total East	Original	Central East	-48.92	2,109.89	2,158.82
Shoulder light load	Original	UPNY-ConEd	3,945.62	2,581.78	-1,363.83
Low Total East	GL	UPNY-ConEd	1,250.58	2,590.35	1,339.77
Shoulder light load	Original	Dunwoodie South	3,173.77	1,876.72	-1,297.05

Caveat

These residuals are diagnostic only because the corresponding OPF runs did not certify feasibility.

Likely Failure Reasons

- ➊ **Voltage and Q infeasibility:** downstate load is moved to proxy buses without matching detailed voltage support.
- ➋ **Load allocation concentration:** zero-load zones G and J rely on proxy weights, causing large new loads at a few buses.
- ➌ **Reactive load modeling:** new proxy load uses simplified Q assumptions, which may over-stress local Q limits.
- ➍ **Mismatch of operating assumptions:** public NYISO load hours do not include matching external interchange, unit commitment, voltage controls, or internal NYISO topology.
- ➎ **Tight branch constraints:** branch 190 and other limits can bind, but they are not the sole blocker.

Recommended Next Diagnostics

- ➊ Run an **eta continuation**: apply 0%, 10%, ..., 100% of the NYISO load-shape shift.
- ➋ At the first failed eta, report:
 - voltage-bound buses,
 - generator Q-limit activity,
 - binding or overloaded branches,
 - zone-level load movement.
- ➌ Audit proxy weights for Zones G, J, and K before fitting X_{GL} .
- ➍ Test alternative reactive-load assumptions for zero-load proxy zones.
- ➎ Only resume Gilboa-Leeds X calibration after at least one public scenario is ACOPF-feasible.

Artifacts Produced

File	Purpose
nyiso_public_scenarios.csv	Six public operating-hour definitions and scale factors.
ny_zonal_load_targets.csv	Scaled A-K zonal MW targets, including S0 and public scenarios.
nyiso_public_interface_targets.csv	Scaled P-32 flow and aggregate-limit references.
nyiso_public_acopf_results.csv	IPOPT ACOPF summary by scenario and topology.
nyiso_public_interface_residuals.csv	Per-interface residuals against public targets.
npcc_original_bus_zone_loads.csv	Bus number, bus name, mapped zone, and original base-case load.
nyiso_public_calibration_results.zip	Packaged calibration result tables.

Current Decision Point

Do not calibrate X_{GL} yet

The public scenario ACOPF base points are not certified feasible, so fitting Gilboa-Leeds against those residuals would tune against failed solutions.

Recommended next step

Use eta continuation and voltage/Q diagnostics to find the smallest public-load shift that breaks the reduced case. Then repair the load allocation or voltage/reactive support assumptions before re-running the grid search.